

Lecture 18

ML-DSA (FIPS 204)

SEC 595: Encrypted Computing

Roadmap

ML-KEM (Lecture 17) establishes a key but authenticates nobody, and today's certificates are signed with RSA or ECDSA, both broken by Shor's algorithm. This lecture builds the lattice replacement, which can be stated in one sentence:

ML-DSA is a Fiat-Shamir transformed Sigma protocol over module lattices, with an abort step.

Section 1 builds the frame on Schnorr's protocol: commitments, Sigma protocols and the Fiat-Shamir transform. Section 2 instantiates it over modules and proves the verification identity. Section 3 names the two hard problems. Section 4 explains the abort, the one step with no analogue in Schnorr. Section 5 shrinks the public key with high bits and hints, and Section 6 gives the parameter sets, their sizes and the complete NTT at $q = 8380417$.

1. Sigma Protocols and Fiat-Shamir

1.1 Commitments

Definition 1 (Commitment scheme). $\text{Com}(m; r)$ produces a commitment c from a message m and randomness r ; an opening reveals (m, r) . The scheme is **hiding** if c reveals nothing about m , and **binding** if c cannot be opened to two different messages.

A commitment is a sealed envelope. The standard example is **Pedersen**: in a group of prime order p with generators g, h , set $\text{Com}(m; r) = g^m h^r$. A collision $g^m h^r = g^{m'} h^{r'}$ with $m \neq m'$ gives $\log_g h = (m - m')(r' - r)^{-1} \bmod p$, so binding rests on the discrete logarithm problem (Lecture 1).

1.2 Sigma Protocols

Definition 2 (Sigma protocol). A three-move protocol for a relation $\mathcal{R}$: the prover sends a **commitment** a , the verifier replies with a uniformly random **challenge** $c \in \mathcal{C}$, and the prover sends a **response** z . It must satisfy

- (i) **completeness**: an honest prover with a witness always convinces the verifier;
- (ii) **special soundness**: from accepting (a, c, z) and (a, c', z') with $c \neq c'$ one can efficiently extract a witness;
- (iii) **honest-verifier zero-knowledge (HVZK)**: a simulator with no witness outputs accepting (a, c, z) with the correct distribution.

By (ii), a prover without a witness can answer at most one challenge for each commitment, so it succeeds with probability at most $1/|\mathcal{C}|$. The simulator in (iii) works out of order: it chooses z first

and solves for a , which a real prover cannot do, since it must send a before seeing c .

1.3 Schnorr Identification

The statement is $h = g^x$ in a group of prime order p ; the witness is x .

1. *Commit*: choose a random k and send $a = g^k$.
2. *Challenge*: the verifier sends a random $c \in \mathbb{Z}_p$.
3. *Respond*: send $z = k + cx \bmod p$.

The verifier accepts iff $g^z = ah^c$.

Proposition 1. *Schnorr's protocol is a Sigma protocol.*

Proof. Completeness: $g^{k+cx} = g^k(g^x)^c$. Special soundness: from $g^z = ah^c$ and $g^{z'} = ah^{c'}$, $g^{z-z'} = h^{c-c'}$, so $x = (z - z')(c - c')^{-1} \bmod p$. HVZK: choose c, z uniformly and set $a = g^z h^{-c}$; this triple has the same distribution as a real one. $\square$

Example 1 (Schnorr, both directions). Work in $\mathbb{F}_{23}^\times$ with $g = 2$, which has order 11 since $2^{11} = 2048 = 89 \cdot 23 + 1$. Take $x = 7$, so $h = 2^7 = 128 \equiv 13$. Exponents are reduced modulo 11.

(a) *Transcripts.* With $k = 3$, $a = 2^3 = 8$. At $c = 2$: $z = 3 + 14 = 17 \equiv 6$, and $g^6 = 64 \equiv 18$ while $ah^2 = 8 \cdot 8 = 64 \equiv 18$. $\checkmark$ At $c = 5$: $z = 3 + 35 = 38 \equiv 5$, and $g^5 = 32 \equiv 9$ while $ah^5 = 8 \cdot 4 = 32 \equiv 9$. $\checkmark$

(b) *Extraction.* $x = (6 - 5)(2 - 5)^{-1} = 8^{-1} \bmod 11 = 7$, since $8 \cdot 7 = 56 \equiv 1$. $\checkmark$

(c) *Simulation.* Fix $c = 5$ and choose $z = 9$ first. Then $g^9 = 512 \equiv 6$, $h^5 \equiv 4$, $4^{-1} \equiv 6 \pmod{23}$, and $a = 6 \cdot 6 = 36 \equiv 13$. Check: $ah^5 = 13 \cdot 4 = 52 \equiv 6 = g^9$. $\checkmark$ The response was chosen before the commitment, and no witness was used.

1.4 Fiat–Shamir

Definition 3 (Random oracle; Fiat–Shamir transform). In the **random oracle model** a hash function H is modelled as a truly random function reachable only by queries. The **Fiat–Shamir transform** replaces the verifier's challenge by $c = H(a \parallel \text{ctx})$, where ctx holds the statement and the message. The pair (a, z) , or equivalently (c, z) , is a **signature**.

Two rules: the statement must be inside the hash, and the challenge space must be too large to guess before committing. Otherwise a forger guesses c , runs the simulator, and outputs a valid signature.

Example 2 (Fiat–Shamir applied to Schnorr). Keep the group of Example 1, with the toy hash $H(a \parallel m) = (3a + 5m) \bmod 11$.

(a) *Sign* $m = 3$ with nonce $k = 3$: $a = 8$, $c = (24 + 15) \bmod 11 = 39 \bmod 11 = 6$, and $z = 3 + 6 \cdot 7 = 45 \equiv 1 \pmod{11}$. The signature is $(c, z) = (6, 1)$.

(b) *Verify*: $a' = g^z h^{-c} = 2 \cdot (13^6)^{-1}$. Here $13^6 \equiv 6$ and $6^{-1} \equiv 4 \pmod{23}$, so $a' = 8$, and $H(8 \parallel 3) = 6 = c$. $\checkmark$

(c) *Another message*: $H(8 \parallel 4) = 44 \bmod 11 = 0 \neq 6$, so the signature does not verify on $m = 4$.

2. The Scheme

2.1 The Ring and the Parameters

Every parameter set uses

$$n = 256, \quad q = 8380417 = 2^{23} - 2^{13} + 1, \quad R_q = \mathbb{Z}_q[X]/(X^{256} + 1),$$

the same ring shape as ML-KEM with a larger modulus. Unlike ML-KEM, seven parameters move between sets: (k, ℓ) , η , τ , γ_1 , γ_2 , ω (Section 6). The bound $\beta = \tau\eta$ is derived; it is not the BKZ block size of Lecture 17.

2.2 KeyGen

$$\mathbf{A} \in R_q^{k \times \ell}, \quad \mathbf{s}_1 \in R_q^\ell, \quad \mathbf{s}_2 \in R_q^k, \quad \mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2.$$

The matrix $\mathbf{A}$ is uniform, expanded from a 32-byte seed ρ . The secrets are short: every coefficient of $\mathbf{s}_1, \mathbf{s}_2$ is uniform on $\{-\eta, \dots, \eta\}$ with $\eta \in \{2, 4\}$. Two differences from ML-KEM: $\mathbf{A}$ is **rectangular**, and the noise is uniform on a box rather than a centred binomial. The pair $(\mathbf{A}, \mathbf{t})$ is an MLWE sample (Lecture 6, §2.3); here $\mathbf{s}_2$ is part of the secret key.

2.3 Sign: Commit and Challenge

Commit. Sample $\mathbf{y} \in R_q^\ell$ with coefficients uniform on $(-\gamma_1, \gamma_1]$ and compute

$$\mathbf{w} = \mathbf{A}\mathbf{y}, \quad \mathbf{w}_1 = \text{HighBits}(\mathbf{w}).$$

HighBits is defined in Section 5; read it for now as “divide by $2\gamma_2$ and round”.

Challenge. Let B_τ be the set of polynomials with exactly τ coefficients in $\{\pm 1\}$ and the rest 0. Then

$$\tilde{c} = H(\mu \parallel \mathbf{w}_1) \in \{0, 1\}^{2\lambda}, \quad c = \text{SampleInBall}(\tilde{c}) \in B_\tau,$$

where μ is a digest of the public key and the message. This is Fiat–Shamir. The challenge space has size $|B_\tau| = \binom{256}{\tau} 2^\tau$, which is $2^{192.8}$, $2^{225.3}$ and $2^{257.0}$ for $\tau = 39, 49, 60$, above 2^λ for $\lambda = 128, 192, 256$.

2.4 Sign: Respond

$$\mathbf{z} = \mathbf{y} + c\mathbf{s}_1 \in R_q^\ell, \quad \sigma = (\tilde{c}, \mathbf{z}, \mathbf{h}),$$

where $\mathbf{h}$ is the hint of Section 5. The commitment $\mathbf{w}_1$ is not sent: the verifier recomputes it and checks that hashing it reproduces $\tilde{c}$.

2.5 The Verification Identity

Proposition 2. *With $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$ and $\mathbf{z} = \mathbf{y} + c\mathbf{s}_1$,*

$$\mathbf{A}\mathbf{z} - c\mathbf{t} = \mathbf{A}\mathbf{y} - c\mathbf{s}_2 = \mathbf{w} - c\mathbf{s}_2.$$

Proof. $\mathbf{A}\mathbf{z} = \mathbf{A}\mathbf{y} + c\mathbf{A}\mathbf{s}_1$ and $c\mathbf{t} = c\mathbf{A}\mathbf{s}_1 + c\mathbf{s}_2$; the term $c\mathbf{A}\mathbf{s}_1$ cancels. □

From public data the verifier recovers the secret commitment $\mathbf{w}$, perturbed by $c\mathbf{s}_2$. If the perturbation cannot change the high bits, then $\text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t}) = \mathbf{w}_1$.

Proposition 3. For $c \in B_\tau$ and $\|\mathbf{s}\|_\infty \leq \eta$, $\|c\mathbf{s}\|_\infty \leq \tau\eta = \beta$.

Proof. Each coefficient of the negacyclic product $c\mathbf{s}$ is a sum of τ terms $\pm s_j$, one for each nonzero coefficient of c , each of absolute value at most η . $\square$

The generic bound $\|ab\|_\infty \leq n\|a\|_\infty\|b\|_\infty$ (Lecture 3) gives $256 \cdot 1 \cdot 2 = 512$ for ML-DSA-44; the true bound is $\beta = 78$. Sparsity buys the factor. The one bound does two jobs: $c\mathbf{s}_2$ is what the verifier must tolerate, and $c\mathbf{s}_1$ is what the signer must hide.

2.6 Schnorr Against ML-DSA

	Schnorr	ML-DSA
statement	$h = g^x$	$\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$
witness	x	$(\mathbf{s}_1, \mathbf{s}_2)$, short
commitment	g^k , k uniform	$\mathbf{w}_1 = \text{HighBits}(\mathbf{A}\mathbf{y})$, $\mathbf{y}$ short
challenge	uniform in $\mathbb{Z}_p$	$H(\mu\ \mathbf{w}_1)$, sparse ± 1
response	$k + cx$	$\mathbf{z} = \mathbf{y} + c\mathbf{s}_1$
verification	$g^z = a h^c$	$\text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t}) = \mathbf{w}_1$
abort	none	unless $\ \mathbf{z}\ _\infty < \gamma_1 - \beta$
secrecy	DLog	MLWE
unforgeability	DLog	MSIS

Every row except the abort is a translation.

3. Security: MLWE and MSIS

3.1 MLWE Hides the Key

The public key is an MLWE sample $(\mathbf{A}, \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2)$. Distinguishing it from uniform, or recovering $\mathbf{s}_1$, breaks MLWE — the same assumption that protects ML-KEM’s public key.

3.2 MSIS Prevents Forgery

Definition 4 (Module-SIS (Lecture 6, §2.4)). Given uniform $\mathbf{A} \in R_q^{k \times \ell}$ and a bound γ , find a nonzero $\mathbf{u}$ with $[\mathbf{A} \mid \mathbf{I}]\mathbf{u} = \mathbf{0}$ and $\|\mathbf{u}\|_\infty \leq \gamma$.

SIS is SVP on the q -ary kernel lattice (Lecture 5, §6). Special soundness gives the connection. Suppose a forger produces two accepting transcripts $(\mathbf{w}_1, c, \mathbf{z})$ and $(\mathbf{w}_1, c', \mathbf{z}')$ with $c \neq c'$. Then

$$\text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t}) = \text{HighBits}(\mathbf{A}\mathbf{z}' - c'\mathbf{t}) = \mathbf{w}_1,$$

so $\mathbf{A}(\mathbf{z} - \mathbf{z}') - (c - c')\mathbf{t}$ is small, being a difference of two elements with the same high bits. Substituting $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$ turns this into a short nonzero $\mathbf{u}$ with $[\mathbf{A} \mid \mathbf{I}]\mathbf{u} = \mathbf{0}$: an MSIS solution.

The formal proof uses a variant, **SelfTargetMSIS**, that folds the hash into the problem, since a forger chooses the message and hence part of the instance. Both the MLWE and the MSIS estimates are computed, and the claimed category is the smaller.

4. Rejection Sampling

4.1 The Leak

Fix one coefficient. Write $u_i = (c\mathbf{s}_1)_i$, so $|u_i| \leq \beta$, and let y_i be uniform on the $2\gamma_1$ integers of $(-\gamma_1, \gamma_1]$. Then

$$z_i = y_i + u_i \text{ is uniform on } (-\gamma_1 + u_i, \gamma_1 + u_i].$$

The shape is fixed, but the **position moves with the secret**.

- *One signature can pin a coefficient:* $z_i = \gamma_1 + \beta$ is possible only if $u_i = \beta$.
- *Many signatures recover the key:* $\mathbb{E}[z_i] = u_i$, which is linear in $\mathbf{s}_1$ through a known c . Averaging over signatures gives a linear system for $\mathbf{s}_1$.

This is an attack, not a gap in a proof: the original NTRUSign was broken this way.

4.2 The Fix

Definition 5 (Rejection condition). Accept $\mathbf{z} = \mathbf{y} + c\mathbf{s}_1$ only if

$$\|\mathbf{z}\|_\infty < \gamma_1 - \beta.$$

Otherwise discard $\mathbf{y}$, $\mathbf{w}$, c and $\mathbf{z}$ *entirely* and restart with a fresh $\mathbf{y}$.

Nothing from a rejected attempt may survive; a partial reuse leaks through the correlation between attempts (Exercise 7). This is **Fiat–Shamir with aborts** (Lyubashevsky).

Proposition 4. Fix $|u| \leq \beta$, let y be uniform on $(-\gamma_1, \gamma_1]$, and put $z = y + u$. Conditioned on $|z| < \gamma_1 - \beta$, z is uniform on the $2(\gamma_1 - \beta) - 1$ integers of that interval, independent of u .

Proof. For $|v| < \gamma_1 - \beta$, the unique y with $z = v$ is $y = v - u$, and $|v - u| < (\gamma_1 - \beta) + \beta = \gamma_1$, so y is in range and is drawn with probability $1/(2\gamma_1)$. Every accepted value has the same probability, whatever u is. $\square$

An accepted $\mathbf{z}$ therefore carries *zero* information about $\mathbf{s}_1$ — not negligible, none. The probability that one coefficient passes is $(2(\gamma_1 - \beta) - 1)/(2\gamma_1)$.

4.3 The Abort Is the Simulator

Proposition 4 gives the HVZK simulator: given c , sample $\mathbf{z}$ uniformly from $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$, set $\mathbf{w}_1 = \text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t})$, and output $(\mathbf{w}_1, c, \mathbf{z})$. This has the same shape as Schnorr’s simulator: z first.

	Schnorr	ML-DSA
mask lives in	the group $\mathbb{Z}_p$	a box $\ \mathbf{y}\ _\infty \leq \gamma_1$
shift by the secret	a bijection	moves the box
repair	none needed	reject and restart

Why not take $\mathbf{y}$ uniform on R_q , as Schnorr does? Because verification is meaningful only for short $\mathbf{z}$: an unbounded $\mathbf{z}$ would satisfy $\text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t}) = \mathbf{w}_1$ trivially, and the MSIS argument of Section 3.2 would collapse.

4.4 An Abort, Traced

The real parameters cannot be written out, so we use the same structure in $\mathbb{Z}_{257}[X]/(X^4 + 1)$ with $k = \ell = 1$, $\eta = 1$, $\tau = 2$, hence $\beta = 2$, and $\gamma_1 = 8$. A polynomial $a_0 + a_1X + a_2X^2 + a_3X^3$ is written (a_0, a_1, a_2, a_3) , with centred coefficients. The mask is uniform on $(-\gamma_1, \gamma_1] = \{-7, \dots, 8\}$, which has 16 values, and the acceptance condition is $\|\mathbf{z}\|_\infty < \gamma_1 - \beta = 6$, i.e. $\|\mathbf{z}\|_\infty \leq 5$.

Example 3 (An abort trace). Take

$$\mathbf{A} = (3, 1, 4, 1), \quad c = (1, 0, -1, 0), \quad \mathbf{s}_1 = (1, -1, 0, 1), \quad \mathbf{s}_2 = (0, 1, -1, 0).$$

(a) *Keys and $c\mathbf{s}_1$.* Negacyclic multiplication gives $\mathbf{A}\mathbf{s}_1 = (3, -6, 2, 0)$, so $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2 = (3, -5, 1, 0)$. Since $c = 1 - X^2$,

$$c\mathbf{s}_1 = \mathbf{s}_1 - X^2\mathbf{s}_1 = (1 - X + X^3) - (X^2 - X^3 - X) = 1 - X^2 + 2X^3 = (1, 0, -1, 2),$$

using $X^5 = -X$. So $\|c\mathbf{s}_1\|_\infty = 2 = \tau\eta$: the bound of Proposition 3 is attained.

(b) *Attempt one* draws $\mathbf{y} = (7, -6, 5, -3)$:

$$\mathbf{z} = (7 + 1, -6 + 0, 5 - 1, -3 + 2) = (8, -6, 4, -1), \quad \|\mathbf{z}\|_\infty = 8 \geq 6.$$

Rejected; everything is discarded.

(c) *Attempt two* draws $\mathbf{y} = (2, -3, 4, 0)$:

$$\mathbf{z} = (2 + 1, -3 + 0, 4 - 1, 0 + 2) = (3, -3, 3, 2), \quad \|\mathbf{z}\|_\infty = 3 < 6.$$

Accepted.

(d) *The identity.* For the accepted attempt, $\mathbf{w} = \mathbf{A}\mathbf{y} = (-7, -11, 17, -6)$ and $\mathbf{A}\mathbf{z} - c\mathbf{t} = (-6, -12, 18, -5)$. Their difference is $(1, -1, 1, 1) = -c\mathbf{s}_2$, since $c\mathbf{s}_2 = (-1, 1, -1, -1)$, as Proposition 2 predicts.

Example 4 (What the rejected $\mathbf{z}$ would have leaked). Continue Example 3: y_i is uniform on the 16 values $\{-7, \dots, 8\}$ and $u_i = (c\mathbf{s}_1)_i \in \{-2, \dots, 2\}$.

(a) *Ranges.* $z_i = y_i + u_i$ ranges over $[-7 + u_i, 8 + u_i]$:

u_i	range of z_i	$z_i = 8$ possible?	surviving z_i ($ z_i \leq 5$)
-2	$[-9, 6]$	no	$[-5, 5]$, 11 values
-1	$[-8, 7]$	no	$[-5, 5]$, 11 values
0	$[-7, 8]$	yes	$[-5, 5]$, 11 values
+1	$[-6, 9]$	yes	$[-5, 5]$, 11 values
+2	$[-5, 10]$	yes	$[-5, 5]$, 11 values

(b) *The rejected candidate* had $z_0 = 8$. Publishing it would rule out $u_0 \in \{-2, -1\}$ (the true value is $u_0 = 1$). With c known, this cuts the possibilities for $\mathbf{s}_1$; over many signatures the key falls out.

(c) *The accepted candidates.* For every u_i , exactly 11 of the 16 values of y_i survive $|z_i| \leq 5$, namely $y_i = v - u_i$ for $v \in [-5, 5]$, and the surviving z_i are uniform on $[-5, 5]$.

(d) *Meaning.* Nothing in an accepted coefficient distinguishes $u_i = 2$ from $u_i = -2$. This is Proposition 4 with $2(\gamma_1 - \beta) - 1 = 11$ values and acceptance probability $11/16 = (2(\gamma_1 - \beta) - 1)/(2\gamma_1)$.

4.5 Three Rejection Tests and the Acceptance Rate

FIPS 204 rejects on three conditions.

1. $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$: hides $\mathbf{s}_1$.
2. $\|\mathbf{r}_0\|_\infty < \gamma_2 - \beta$, where $\mathbf{r}_0 = \text{LowBits}(\mathbf{w} - c\mathbf{s}_2)$. If the low part of $\mathbf{w} - c\mathbf{s}_2$ stays β away from a boundary, then $\text{HighBits}(\mathbf{w} - c\mathbf{s}_2) = \text{HighBits}(\mathbf{w}) = \mathbf{w}_1$ and the verifier agrees; and whether $\mathbf{w}$ sits near a boundary would otherwise leak $\mathbf{s}_2$.
3. $\|c\mathbf{t}_0\|_\infty < \gamma_2$ and $\mathbf{h}$ has at most ω ones (Section 5).

Treating the 256ℓ coefficients of $\mathbf{z}$ and the $256k$ of $\mathbf{r}_0$ as independent, Proposition 4 and its analogue for $\mathbf{r}_0$ give

$$\Pr[\text{accept}] \approx \left(\frac{2(\gamma_1 - \beta) - 1}{2\gamma_1} \right)^{256\ell} \left(\frac{2(\gamma_2 - \beta) - 1}{2\gamma_2} \right)^{256k} \approx \exp\left(-256\beta \left(\frac{\ell}{\gamma_1} + \frac{k}{\gamma_2} \right) \right).$$

Set	β	$\Pr[\mathbf{z}]$	$\Pr[\mathbf{r}_0]$	Both	Attempts
ML-DSA-44	78	0.541	0.430	0.233	4.30
ML-DSA-65	196	0.619	0.316	0.195	5.12
ML-DSA-87	120	0.662	0.390	0.258	3.87

The last column is $1/\Pr[\text{accept}]$; the exponential approximation gives the often-quoted 4.25, 5.09 and 3.85. The low-bits test dominates because $\gamma_2 \ll \gamma_1$. ML-DSA-65, not ML-DSA-87, is the slowest to sign: $\eta = 4$ pushes β to 196.

Signing time is variable, and this is safe: the loop count depends on the fresh $\mathbf{y}$ and the public μ . The rejection test itself must be constant-time, since *which* coefficient failed depends on the secret.

5. High Bits and Hints

5.1 Two Decompositions

Definition 6 (Power2Round and Decompose). For $r \in \mathbb{Z}_q$:

- $\text{Power2Round}(r) = (r_1, r_0)$ with $r = r_1 2^{13} + r_0$ and $-2^{12} < r_0 \leq 2^{12}$. Here r_1 takes $2^{23-13} = 2^{10}$ values.
- $\text{Decompose}(r) = (r_1, r_0)$ with $r = r_1(2\gamma_2) + r_0$ and $|r_0| \leq \gamma_2$, where $2\gamma_2 \mid q - 1$; then $\text{HighBits}(r) = r_1$ and $\text{LowBits}(r) = r_0$. Here r_1 takes $(q - 1)/(2\gamma_2) = 44$ or 16 values.

Power2Round shrinks the public key; Decompose shrinks the commitment. A ciphertext need only decrypt, so ML-KEM's compression can absorb rounding error into the noise. A signature must verify *exactly*, so discarded bits must be reconstructed. That is the role of the hint.

5.2 The Public Key Carries Only $\mathbf{t}_1$

Apply Power2Round to $\mathbf{t}$: $\mathbf{t} = \mathbf{t}_1 2^{13} + \mathbf{t}_0$. The public key is $(\rho, \mathbf{t}_1)$ and $\mathbf{t}_0$ goes into sk . A coefficient of $\mathbf{t}$ needs 23 bits, one of $\mathbf{t}_1$ needs 10, so a ring element shrinks from 736 to 320 bytes:

Set	pk with full $\mathbf{t}$	pk with $\mathbf{t}_1$
ML-DSA-44	$32 + 4 \cdot 736 = 2976$	$32 + 4 \cdot 320 = 1312$
ML-DSA-65	$32 + 6 \cdot 736 = 4448$	$32 + 6 \cdot 320 = 1952$
ML-DSA-87	$32 + 8 \cdot 736 = 5920$	$32 + 8 \cdot 320 = 2592$

About 56% is saved. But the verifier can now only compute

$$\mathbf{A}\mathbf{z} - c\mathbf{t}_1 2^{13} = (\mathbf{A}\mathbf{z} - c\mathbf{t}) + c\mathbf{t}_0 = (\mathbf{w} - c\mathbf{s}_2) + c\mathbf{t}_0,$$

and $\|c\mathbf{t}_0\|_\infty$ can reach $\tau \cdot 2^{12} = 39 \cdot 4096 = 159744 > \gamma_2 = 95232$ for ML-DSA-44, enough to cross a high-bits boundary.

5.3 The Hint

Definition 7 (Hint). $\mathbf{h} \in \{0, 1\}^{256k}$, with $h_i = 1$ exactly where adding $c\mathbf{t}_0$ changes the high bits: $\text{HighBits}((\mathbf{w} - c\mathbf{s}_2)_i) \neq \text{HighBits}((\mathbf{w} - c\mathbf{s}_2 + c\mathbf{t}_0)_i)$. The signer builds $\mathbf{h}$ with `MakeHint`; the verifier corrects with `UseHint`.

After test 3, $\|c\mathbf{t}_0\|_\infty < \gamma_2$, so a crossing moves the high part by exactly ± 1 , in a direction the verifier can read from the sign of the low part. One bit per coefficient suffices: the hint says *whether*, not *which way*. Since $\mathbf{h}$ has at most ω ones, it is encoded as ω index bytes plus k cumulative-count bytes, $\omega + k$ bytes in all.

5.4 Verify

1. Recompute μ from the public key and the message.
2. $c = \text{SampleInBall}(\tilde{c})$.
3. $\mathbf{w}'_1 = \text{UseHint}(\mathbf{h}, \mathbf{A}\mathbf{z} - c\mathbf{t}_1 2^{13})$.
4. Accept iff $\tilde{c} = H(\mu \parallel \mathbf{w}'_1)$, $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$, and $\mathbf{h}$ has at most ω ones.

There is no loop: verification is straight-line.

6. Parameters and the NTT

6.1 Parameter Sets

Set	(k, ℓ)	η	τ	β	γ_1	γ_2	ω	λ	pk	sk	σ	Cat.
ML-DSA-44	(4, 4)	2	39	78	2^{17}	$(q-1)/88$	80	128	1312	2560	2420	2
ML-DSA-65	(6, 5)	4	49	196	2^{19}	$(q-1)/32$	55	192	1952	4032	3309	3
ML-DSA-87	(8, 7)	2	60	120	2^{19}	$(q-1)/32$	75	256	2592	4896	4627	5

Sizes are in bytes; $\gamma_2 = 95232$ for ML-DSA-44 and 261888 for the other two. The name is the shape of $\mathbf{A}$. There is no category-1 set. The ranks k and ℓ tune the MLWE and MSIS estimates separately (core-SVP, Lecture 17), which is why $\mathbf{A}$ is rectangular.

6.2 Sizes

Proposition 5.

$$pk = 32 + 320k, \quad \sigma = \frac{\lambda}{4} + 32\ell(1 + \log_2 \gamma_1) + \omega + k \quad \text{bytes.}$$

Proof. $pk = (\rho, \mathbf{t}_1)$: 32 bytes of seed and k ring elements of 256 coefficients of 10 bits, i.e. 320 bytes each. In σ , $\tilde{c}$ has 2λ bits = $\lambda/4$ bytes; each coefficient of $\mathbf{z}$ satisfies $|z_i| < \gamma_1$ and packs into $1 + \log_2 \gamma_1$ bits, giving $256(1 + \log_2 \gamma_1)/8$ bytes per ring element; and $\mathbf{h}$ is $\omega + k$ bytes. $\square$

$$\begin{aligned}
\text{ML-DSA-44: } & 32 + 4 \cdot 576 + (80 + 4) = 32 + 2304 + 84 = 2420, \\
\text{ML-DSA-65: } & 48 + 5 \cdot 640 + (55 + 6) = 48 + 3200 + 61 = 3309, \\
\text{ML-DSA-87: } & 64 + 7 \cdot 640 + (75 + 8) = 64 + 4480 + 83 = 4627.
\end{aligned}$$

For ML-DSA-44, $\mathbf{z}$ is $2304/2420 = 95.2\%$ of the signature: γ_1 is the parameter worth fighting over. (The secret key stores ρ, K, tr in 128 bytes, then $\mathbf{s}_1, \mathbf{s}_2$ at 3 or 4 bits per coefficient and $\mathbf{t}_0$ at 13: $128 + 8 \cdot 96 + 4 \cdot 416 = 2560$ for ML-DSA-44.)

6.3 A Complete NTT at $q = 8380417$

Modulo $X^{256} + 1$ a complete negacyclic NTT needs a primitive 512-th root of unity, i.e. $512 \mid q - 1$ (Lecture 3). Here

$$q - 1 = 8380416 = 2^{13} \cdot 3 \cdot 11 \cdot 31,$$

so the two-adicity is 13. FIPS 204 uses $\zeta = 1753$, with $\zeta^{256} \equiv -1$ and $\zeta^{512} \equiv 1 \pmod{q}$; the first congruence makes the order exactly 512. All eight layers run:

$$R_q \cong \prod_{i=0}^{255} \mathbb{Z}_q[X] / (X - \zeta^{2 \text{brv}(i)+1}) \cong \mathbb{Z}_q^{256},$$

where brv is 8-bit bit reversal. There are 256 linear factors, so multiplication is 256 scalar products.

Example 5 (The two-adicity comparison).

	$q = 3329$ (ML-KEM)	$q = 8380417$ (ML-DSA)
$q - 1$	$3328 = 2^8 \cdot 13$	$8380416 = 2^{13} \cdot 3 \cdot 11 \cdot 31$
two-adicity	8	13
$256 \mid q - 1$?	yes	yes
$512 \mid q - 1$?	no	yes
NTT layers ($n = 256$)	7	8
factors	128 of degree 2	256 of degree 1
pointwise step	4 mults +1 by ζ_i	1 mult
largest complete n	128	4096

For $q = 3329$ the deepest root is of order 256, so splitting stops at $X^2 - \zeta_i$, and $(a_0 + a_1 X)(b_0 + b_1 X) = (a_0 b_0 + \zeta_i a_1 b_1) + (a_0 b_1 + a_1 b_0)X$ costs four multiplications plus one by ζ_i . A complete negacyclic NTT of length n needs $2n \mid q - 1$: $2n \leq 2^8$ gives $n \leq 128$ for 3329, and $2n \leq 2^{13}$ gives $n \leq 4096$ for 8380417. ML-DSA uses 9 of its 13 factors of two.

6.4 Three Divisibility Conditions on q

$$\begin{aligned}
512 \mid q - 1 & \quad \text{the NTT is complete;} \\
2\gamma_2 \mid q - 1 & \quad \text{Decompose has equal intervals;} \\
2^{13} \mid q - 1 & \quad \text{Power2Round has a full top interval.}
\end{aligned}$$

Hence $\gamma_2 = (q - 1)/88$ or $(q - 1)/32$, with $88 = 2^3 \cdot 11$ and $32 = 2^5$, both divisors of $q - 1$ as required. The modulus had to be about 23 bits anyway, to hold γ_1 and the 13 low bits of $\mathbf{t}$, so two-adicity 13 costs nothing.

6.5 Against the Alternatives

Scheme	Construction	pk	sig	Rests on
Ed25519	Fiat–Shamir	32	64	DLog (Shor)
ML-DSA-44	Fiat–Shamir + aborts	1312	2420	MLWE, MSIS
FN-DSA-512	hash-and-sign trapdoor	897	666	NTRU, SIS
SLH-DSA-128s	hypertree of OTS	32	7856	hash security

FN-DSA is smaller but needs constant-time Gaussian sampling and floating-point arithmetic. SLH-DSA is larger but assumes only a hash (Lecture 16). ML-DSA is the general-purpose default.

Summary

1. ML-DSA is Schnorr over a module, made non-interactive by Fiat–Shamir: commit $\mathbf{w}_1 = \text{HighBits}(\mathbf{A}\mathbf{y})$, challenge $c = H(\mu\|\mathbf{w}_1)$, respond $\mathbf{z} = \mathbf{y} + c\mathbf{s}_1$.
2. Verification rests on $\mathbf{A}\mathbf{z} - c\mathbf{t} = \mathbf{w} - c\mathbf{s}_2$ and on $\|c\mathbf{s}\|_\infty \leq \tau\eta = \beta$, which holds because c is sparse.
3. MLWE hides the key; MSIS, via special soundness, prevents forgery.
4. Without the abort, $\mathbf{z}$ leaks $\mathbf{s}_1$. Rejecting unless $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$ makes $\mathbf{z}$ uniform and independent of the secret; this is the HVZK simulator. Signing takes about 4–5 attempts.
5. Power2Round cuts the public key by about 56%, and a hint of $\omega + k$ bytes repairs verification. Since $512 \mid q - 1$, the NTT is complete, with 256 linear factors.

Exercises

1. In the order-11 subgroup of $\mathbb{F}_{23}^\times$ with $g = 2$ and $h = 13$, show that the Pedersen commitments $\text{Com}(3; 4)$ and $\text{Com}(0; 6)$ are equal, and use the collision to compute $\log_g h$.
2. With the group and toy hash of Example 2, sign $m = 5$ with nonce $k = 4$ and verify the signature.
3. For each coefficient of the rejected $\mathbf{z} = (8, -6, 4, -1)$ of Example 3, list the values of $u_i \in \{-2, \dots, 2\}$ that remain possible. Show that no coefficient of the accepted $\mathbf{z} = (3, -3, 3, 2)$ rules out any value, and explain why this follows from Proposition 4.
4. Compare the bound $\beta = \tau\eta$ with the generic bound $n\|c\|_\infty\|s\|_\infty = 256\eta$ for each parameter set. For which set is the ratio largest?
5. If γ_1 for ML-DSA-44 were doubled to 2^{18} , by how many bytes would the signature grow, and what would the expected number of signing attempts become? Is the trade worth it?
6. Verify that $1753^{256} \equiv -1$ and $1753^{512} \equiv 1 \pmod{8380417}$, and explain why the first congruence shows the order is exactly 512. Explain why no element of $\mathbb{Z}_{3329}^\times$ has order 512.
7. An implementer proposes a faster loop: on rejection, keep $\mathbf{y}$ (hence $\mathbf{w}$ and $\mathbf{w}_1$) and redraw only the challenge. Explain why the accepted $\mathbf{z}$ is then no longer independent of $\mathbf{s}_1$, and what an attacker obtains if two transcripts with the same $\mathbf{w}_1$ and different challenges ever become public (compare special soundness).